Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Full-stack engineer with 2+ years shipping **generative AI products, high-performance interfaces, and scalable backend systems**; experienced in LLM orchestration, RAG, data extraction, document processing, vector search, and reliable production delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Architected a reliable multi-tenant generative AI platform using TypeScript, Python, FastAPI, and PostgreSQL, supporting 500+ users while reducing high-visibility dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant LLM orchestration pipelines processing 1,000+ AI jobs and 5,000+ recurring monthly integrations with explicit state transitions, failure isolation, validation, and 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed full-stack generative AI functionality with a responsive React interface and modular Python APIs, evaluating 10,000+ submissions while serving 10,000+ daily requests for 150+ instructors.
- Orchestrated multimodal LLMs and complex OCR data-extraction workflows with concurrent processing and model fallbacks, reducing feedback turnaround 60% while sustaining 90%+ consistency across text and handwritten documents.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built large-scale scalable model-training and data pipelines processing 100,000+ records, improving long-horizon predictions 18% through Python, transformer architectures, feature extraction, learned embeddings, and rigorous reproducible experimentation.
- Designed a spatial-temporal transformer integrating heterogeneous context through multi-head attention, validating 10,000 scenarios and improving system stability 15% through comprehensive rigorous model evaluation and performance analysis.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Shipped cross-platform backend services and data-extraction integrations using Python, Java, AWS Glue, and S3, reducing response times 30%, technical debt 20%, and pipeline overhead across distributed systems.
- Implemented 20+ comprehensive automated tests and global CI/CD workflows across Linux, Windows, and macOS, closely collaborating with Microsoft and GitHub engineers while improving large-scale data-processing efficiency 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed high-performance React features and eight backend services for authentication, resource allocation, billing, permissions, and notifications, synchronizing two million records across locations through REST APIs and WebSockets.
- Containerized full-stack applications and automated testing and reliable deployment through Jenkins, achieving 95%+ build validation while reducing deployment time 50% and code-review turnaround 25% across rapid releases.

Skills

Generative AI: LLM orchestration, RAG, agents, prompt engineering, data extraction, document processing, embeddings, similarity search, model evaluation
Front End: TypeScript, JavaScript, React.js, Next.js, HTML, CSS, TailwindCSS, responsive design, componentization
Backend: Python, Java, FastAPI, Flask, Node.js, Express.js, REST APIs, microservices, distributed systems, concurrency
Data: PostgreSQL, SQL Server, MongoDB, Redis, FAISS, Pinecone, SQL, schema design, query optimization, vector search
Platform: AWS, Azure, GCP, Docker, Kubernetes, CI/CD, Git, Jenkins, Sentry, system testing, Agile, product development

Projects

Big-Brain Live | *RAG, LLaMA-2, FAISS, Pinecone*

- Built a semantic retrieval and document-analysis platform over 50,000+ documents using LLaMA-2, FAISS, and Pinecone, delivering sub-second contextual search, summarization, collaborative workflows, and 95% query accuracy.

CaptionCraft Live | *Python, LLMs, Data Extraction*

- Developed generative AI inference APIs and evaluation pipelines across four models, 10,000 images, and 100,000 captions, improving extraction accuracy from 81% to 96% through systematic testing.

Chat2SQL Code | *Java, TypeScript, AI Search*

- Extended a scalable AI-powered full-stack software development platform supporting 10+ databases, implementing Java backend and TypeScript frontend features, schema-aware retrieval, query validation, and Docker-based cross-platform production delivery.